

<https://doi.org/10.1038/s44271-024-00131-3>

# Infants evaluate informativeness of evidence and predict causal events as revealed in theta oscillations and predictive looking



Katarina Begus<sup>1</sup>✉ & Elizabeth Bonawitz<sup>2</sup>

This study investigates 16-month-old infants' sensitivity to the informativeness of evidence and its potential link to infants' ability to draw accurate causal inferences and predict unfolding events. Employing concurrent EEG and eye tracking, data from 66 infants revealed significantly increased theta oscillatory activity when infants expected to see causally unconfounded evidence compared to confounded evidence, suggesting heightened cognitive engagement in anticipation of informative evidence. Crucially, this difference was more pronounced in the subset of infants who later made correct predictions, suggesting that they had correctly inferred the causal structure based on the evidence presented. This research sheds light on infants' motivation to seek explanatory causal information, suggesting that even at 16 months, infants can strategically direct attention to situations conducive to acquiring informative evidence, potentially laying the groundwork for the impressive abilities of humans to rapidly acquire knowledge and develop causal theories of the world.

Living organisms are faced with a constant challenge: how to obtain and evaluate information needed to achieve mastery of the environment, while rationally investing one's limited cognitive resources. This challenge is arguably most pronounced in human infants, who are born into a socio-cultural world presenting them with vast amounts of abstract knowledge systems and opaque causal relationships<sup>1</sup>, while their cognitive capacities are far less developed compared to the mature members of their species. Recent evidence from developmental research has shown that human infants appear well equipped to deal with the seemingly insurmountable amount of information through specialized learning mechanisms that allow rapid and systematic accumulation of information<sup>2-4</sup>. However, there is less research on whether infants possess the foresight to anticipate whether certain events have the potential to offer explanatory clarity, especially in the context of causal reasoning. The ability to discern which events are truly informative, and to distinguish spurious correlations from unambiguous causal evidence, is essential for optimal learning, yet it is poorly understood in early development. In this paper, we introduce a paradigm that delves into the hypothesis that infants are sensitive to events that are likely to yield informative outcomes, and that this sensitivity may allow human infants to tackle one of the biggest challenges of learning about their world - inferring opaque causal structures.

Despite having little control over the environment, and limited means of gathering information, human infants possess several cognitive mechanisms and biases that have been identified as guiding and facilitating their learning. For example, infants have been shown to have remarkable domain general learning mechanisms, which guide attention to novelty<sup>5</sup> and surprising and belief-violating events<sup>6,7</sup>. These mechanisms help direct infants' attention to data in which there is something to be learned (either because it is new or because something about the infants' beliefs requires updating). Other domain general learning mechanisms direct infants' attention towards stimuli that are neither too predictable nor fully unpredictable<sup>8,9</sup>, thereby maximising the likelihood of learning progress. Furthermore, infants also have specialized social learning abilities, leading them to pay special attention to the most prominent source of information in their world - other people. They pay special attention to faces<sup>10</sup> and speech<sup>11</sup>, particularly when these come with direct gaze<sup>12</sup>, or at a pitch and rhythm specifically directed at infants as the recipients<sup>13</sup>. Doing so enables infants to take full advantage of the information transmitted through direct communication from others<sup>14</sup>. Moreover, learning from social agents plays a privileged role in supporting particularly rapid and accurate inferences<sup>15</sup>. Infants not only attend to the social agents per se, but seem to prioritize encoding events that are the targets of others' attention<sup>16</sup>, and thus offload the selection process of what is to be learned on more mature social agents.

<sup>1</sup>University of Copenhagen, Copenhagen, Denmark. <sup>2</sup>Harvard Graduate School of Education, Cambridge, MA, USA. ✉e-mail: [khb@psy.ku.dk](mailto:khb@psy.ku.dk)

This expectation of learning from social agents and modulation of attention becomes selective early in life, with infants showing abilities to discern who or what is most informative<sup>17–21</sup>, providing initial clues that infants have mechanisms that allow them to predict whether future events will be worthy of their limited attention.

While the above research demonstrates impressive abilities of infants to select information and learn efficiently, much of this work on infant learning and attention depicts cases in which infants are *reacting* to events and using such violations of expectations, prediction errors, and social cues to guide attention to continuing and future events. But there are many situations in which events have not yet come to pass but one may benefit from evaluating whether the anticipated event or piece of evidence is likely to be informative or not. For instance, imagine you are trying to find out which water tap gives you warm water and which cold; or which switch turns on a light, while you do not have the possibility to test out the switches yourself. When observing others interacting with these switches, it would be rational to attend to events in which individual switches are flipped and the light changes state. At the same time, you would be wise to avoid forming any conclusions based on events in which multiple switches are flipped simultaneously, because the evidence would be inconclusive. Similarly, in order to establish what a novel word refers to, it would be optimal to isolate potential referents, and avoid learning labels in contexts of referential ambiguity. Thus, if one is able to discern which events are truly informative, and which ones provide only temporal contingencies without disambiguating information, one can avoid false inferences. Human culture is full of such ambiguities and opaque causal relationships. If infants are to rapidly and efficiently acquire knowledge, it would be highly beneficial to be sensitive to the informativeness of evidence that they observe, or expect to observe, and selectively harness only the information that offers explanatory clarity.

### Approach: Evaluation of evidential informativeness in infancy

How can one approach the question of whether infants or even children are able to evaluate the informativeness of evidence? Traditionally, studies have been restricted to behavioral measures. For example, preschool-aged children have been shown to prefer exploring the same toy when the evidence about its functions had been confounded (i.e. multiple simultaneous manipulations resulting in multiple simultaneous effects), but preferred to explore a novel toy when the evidence they had observed had been unconfounded and informative (individual manipulations and the resulting effects had been demonstrated sequentially)<sup>22</sup>. Children's exploration in this study was presumably motivated by the desire to disambiguate the causal mechanism of the toy, a motivation which was absent in the case where unconfounded evidence was presented. This study suggests both that children are motivated to seek causal explanations of the world, as well as that children are able to evaluate the evidence they observe and seek further information only when the evidence is unsatisfying. But one may not always have the opportunity to perform interventions and disambiguate data themselves. It may therefore be a good strategy to evaluate the informativeness of evidence even before it is given and to only engage with information that is informative. Such selectivity would optimize the use of one's cognitive resources and prevent potential unjustified conclusions formed based on confounded information. This may be an even more pressing problem for individuals with limited possibilities of manipulating and performing interventions in the world, such as infants.

We investigate whether infants are sensitive to the informativeness of evidence, specifically whether they distinguish confounded from unconfounded information, and preferentially engage with informative unconfounded information. To do this, we first familiarize infants with a causal structure, and then present them with situations in which a function of a novel object either can be discovered or cannot be discovered because of confounding evidence. We investigate whether infants distinguish between these scenarios and show heightened cognitive engagement in situations where the expected information is unconfounded. We do this by measuring their neural activity, specifically EEG theta oscillations.

### Theta reflecting information expectation

Briefly, theta activity is the oscillatory activity recorded by scalp EEG and spanning roughly between 4 and 8 Hz in human adults<sup>23</sup>, and 3–5 Hz in infants<sup>24–26</sup>. Theta rhythms have been identified as involved in learning in both adults and infants<sup>27</sup>. For example, research in infants has revealed increased theta activity following infants' observation of surprising events<sup>28,29</sup>, during internally-controlled visual attention<sup>30,31</sup> and while observing naturalistic social stimuli<sup>32</sup>, suggesting a link between theta activity and states of heightened attention and cognitive engagement in infants. Moreover, frontal theta activity during infants' active exploration of novel objects has been shown to be predictive of the degree to which infants had encoded the features of the explored objects<sup>36</sup>, demonstrating an analogous relationship between theta activity and learning in infants as has been found in adults<sup>23,33,34</sup>. Finally, theta activity in infants has also been shown in anticipation of receiving information. In a study where infants were faced with two informants, one providing information and the other not, infants showed more theta activity in expectation of receiving information than when no information could be expected<sup>25</sup>. Such selective anticipatory theta activity is proposed to indicate that infants modulate their cognitive engagement, and selectively invest their cognitive resources according to the expected provision of information. Based on this evidence, we utilize this measure of theta to ask whether infants not only anticipate and prepare for learning when information can be expected, but whether they evaluate the quality of the expected information itself and engage their cognitive resources only when informative evidence is to be presented, that is when the expected information is unconfounded.

### Predictive saccades

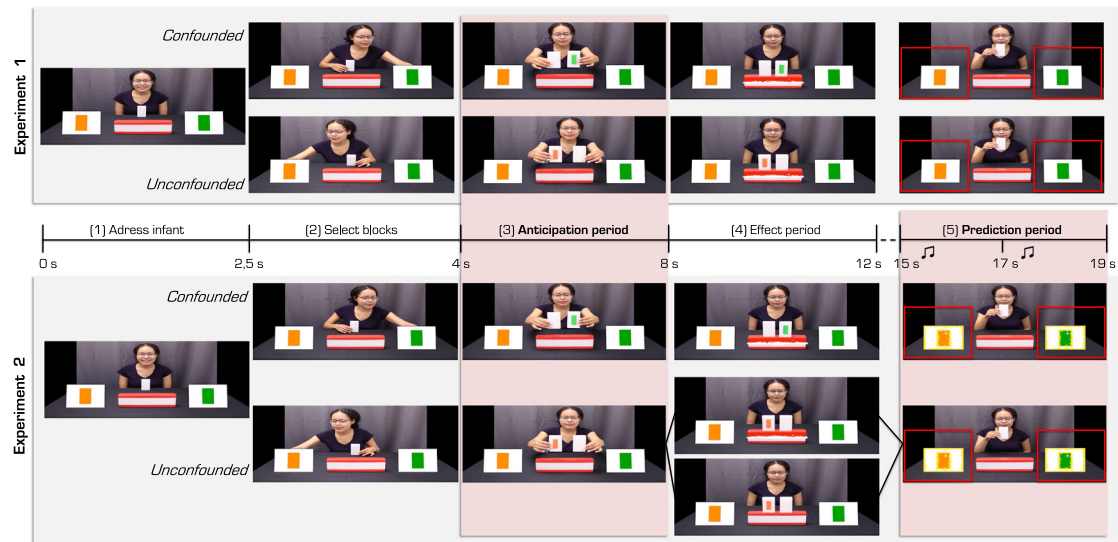
Importantly, we not only measured infants' neural activity in anticipation of confounded or unconfounded information, but also investigated behavioral markers that would indicate infants' understanding of the presented causal structure. To do this, we concurrently recorded eye-tracking data and investigated whether infants would make correct predictions, measured by anticipatory saccades, in situations where they received unconfounded information and could therefore infer the causal role of a novel object. This approach, combining both neural activity prior to presentation of evidence, and behavioral predictions that follow the neural activity, enables us to go beyond measuring only observable behavior and inferring the mechanisms that might lead to behavior post-hoc. Instead, we can directly measure modulations of relevant neural activity to investigate what conditions might precede or even support certain behaviors.

We hypothesized that if infants in our paradigm are sensitive to the informativeness of the evidence they are about to observe, they should exhibit more theta activity in anticipation of unconfounded than confounded information. Moreover, we hypothesized that if infants correctly inferred the causal role of the objects when unconfounded data was presented, they might successfully predict events that would follow. Finally, and crucially, we hypothesized there may exist a relation between infants' success in inferring the causal structure, as evidenced in the accuracy of their predictive saccades, and their sensitivity to the informativeness of evidence, expected to be reflected in theta activity.

### Design and Predictions

Two experiments were conducted to investigate these questions. *Experiment 1* successfully elicited the neural activity of interest, however due to methodological details, predictive saccades were rarely observed and therefore this measure was not informative. In *Experiment 2*, the EEG measures remained identical (Fig. 1), but some minor methodological changes were implemented to improve the eyetracking measures. We combine description of the studies below for concision, but maintain some separation in Fig. 1 caption, results, and methods for methodological clarity and analytic transparency.

Both experiments first familiarized infants with an agent placing individual blocks on a machine, which produced an audio-visual effect when in contact with some blocks and not with others. After observing the



**Fig. 1 | Test trial timeline.** Test trial video timelines for *Experiment 1* (top) and *Experiment 2* (bottom). In the depicted example, ‘active’ blocks were sorted into the green box, and ‘inactive’ blocks into the orange box (colours/locations were counterbalanced across infants in the experiments). From left to right: (1) The agent addresses the infant. (2) The agent picks up the unmarked block and a block from one of the sorting boxes. (3) *Anticipation period*: the pair of blocks is held above the machine. (4) *Effect period*: The blocks are placed on the machine and an audio-visual effect is produced (Experiment 1: effect is triggered in all trials; Experiment 2: effect is triggered in 100% of *Confounded* and 50% of *Unconfounded* trials). The agent then

retrieves the blocks from the machine and hides the marked block (12–15 s, not shown in the figure). (5) *Prediction period*: the agent holds the unmarked block centrally above the machine. In Experiment 2 only, both sorting boxes light up with an accompanying sound at the start of prediction period. In both experiments, a second sorting sound is played after 2 s from the beginning of *Prediction* period, but neither of the sorting boxes light up, and the agent remains holding the block in the centre until the end of the trial. *Note*: Panel (5) - red frames surrounding the sorting boxes depict the areas of interest (AOIs) and were not visible on the stimuli presentation screen.

effect (or lack thereof), the individual blocks were sorted into one of two boxes, marked with different colors, one containing ‘active’ and the other ‘inactive’ blocks. In addition to trials with individual blocks, infants saw trials in which different combinations of 2 blocks were placed on the machine at the same time. In pair-block trials, the blocks were taken from the sorting boxes and were marked with the corresponding colors (counterbalanced across participants). The machine produced an effect consistent with the data in the individual-block trials: if the blocks that triggered the effect (active) in individual trials were placed in a (e.g.) green box, then trials with 2 green (active) blocks would likewise trigger the effect, whereas 2 orange (inactive) blocks would not. Additionally, when one of each type of blocks was placed on the machine (active and inactive) the audio-visual effect would always be triggered. This familiarization served to establish a sorting rule (active and inactive blocks are placed in separate boxes) and a causal structure, according to which the presence of at least one of the active blocks on the machine would always lead to the audio-visual effect, whereas inactive blocks alone would not trigger the effect. All infants in the sample saw minimum 1 of each type of familiarisation trials (2 individual, 3 pair trials) before viewing test trials.

In the test trials, a novel unmarked block was placed on the machine together with one of the active or inactive blocks (Fig. 1), creating two types of trials. When a novel block is paired with an active block, an infant can expect to see the audio-visual effect when the blocks are placed on the machine, because the presence of one active block alone is sufficient to trigger the effect. However, in such a case, the functional role of the novel block remains unknown, because the information about the triggering of the effect is *Confounded* by the presence of the active block. Conversely, when a novel block is paired with an inactive one, the infant cannot know whether the effect will be triggered, but they can expect to find out whether the novel block is one of the active or inactive ones: if an effect is triggered, the novel block must be active, if no effect is triggered, the novel block must likewise be inactive. In other words, they can expect information that is causally relevant and *Unconfounded*. At the end of the test trials, the agent hides the known (and visually marked, active or inactive) block, and holds the novel

unmarked block centrally between the two sorting boxes until the end of the trial.

We investigate whether infants are sensitive to the difference between the two types of trials (*Confounded* and *Unconfounded*), by measuring EEG theta oscillations during the *Anticipation* period (Figs. 1–3) of the trials, when the pairs of blocks are held above the machine, prior to placing them on the surface and before seeing whether the effect is triggered or not. We hypothesized that if infants are able to evaluate the informativeness of the expected evidence, they should recognise that only *Unconfounded* trials offer to reveal the causal role of the novel block, and therefore more oscillatory activity in the theta frequency band (3–5 Hz) should be elicited in the *Anticipation* period of *Unconfounded* compared to the *Confounded* information trials. Furthermore, after seeing the effect (or lack thereof) in *Unconfounded* trials, infants may be able to infer the causal role of the novel object, and this may allow them to form predictions about which of the sorting boxes the novel block will subsequently be placed into. We investigate if this is the case with *Experiment 2*, in which infants’ gaze behavior at the end of the trials (*Prediction* period, Fig. 1 – (5)) was analyzed for correct or incorrect predictions (saccades towards the sorting boxes). Critically, we hypothesized that an ability to recognise and distinguish confounded from unconfounded information is crucial for correctly inferring causal relationships. Therefore, we predicted that infants, who succeed in making correct predictions about the causal roles of novel objects, will show greater differences in neural activity between *Unconfounded* and *Confounded* trials.

## Methods

The study was pre-registered using the AsPredicted platform (on 21.05.2021, <https://aspredicted.org/66h9e.pdf>). Minor discrepancies between performed and planned analyses are reported in Supplementary Materials. The study methods were approved by the ethics committee of University of Copenhagen. An informed consent for participation was collected from accompanying caretakers for each of the participating infant, families were not financially compensated but received small gifts for their participation.

## Participants

A total of 66, 15–17-month-old infants were included in the final sample (39 female (as reported by infants' parents at time of registration to the recruitment database); age range: 460–532 days), 34 in Experiment 1 (20 female); 32 in Experiment 2 (19 female). Ethnicity information was not collected. An additional 58 infants were tested but excluded from analysis due to fussiness (study not completed due to infant's behaviour as judged by the experimenter or accompanying caregiver during the study; 25), equipment failure (5), caregiver interference (1), or insufficient data (27).

## Procedure

Infants were fitted with an EEG sensor-cap (EGI, 128-channels), a bulls-eye sticker for the eyetracker (EyeLink1000), and sat in their caregiver's lap, approximately 60 cm from the presentation screen. Videos were presented on a 34 x 27 cm (1280 x 1024 resolution) screen, eyetracking and EEG data were sampled at 1000 Hz, and a video of the participants was recorded using a webcam placed on the top of the presentation screen. All measures were recorded simultaneously and synchronized online via time-stamped events. Caregivers were instructed not to interact with the infants, with the exception of maintaining the infants' position constant with respect to the screen and eyetracker; and preventing the infants from interfering with the EEG headgear. An attention-grabbing animated video was played until the infant was positioned correctly and the eyetracker detected their eyes, this was followed by a five-point calibration procedure, after which the experiment began. The stimuli were played continuously as long as the infant was paying attention, or until the maximum number of trials. If the infant became upset or inattentive, the experiment was paused and an attention-grabbing video was played, after which the experiment was resumed or the testing session was terminated.

## Stimuli

Videos depicted a centrally positioned female agent, sitting behind a table, facing the camera. Two same-sized (virtual) sorting boxes lay on the left and right edge of the table, in front of the agent, and were marked with orange (left) and green (right) colours. A third box (the machine) lay at the centre of the table, equidistant from the two sorting boxes (see Fig. 1). Each video began with the agent looking directly at the camera, smiling and addressing the infant with "Hi baby, look at this!" (in the local language - Danish). The following sections describe the stimuli as presented in *Experiment 2*. Details that were not included in stimuli of *Experiment 1* are specified below. The sequence of trials was 12 Familiarisation trials (6 individual and 6 pair trials), followed by (maximum) 16 Test trials (interspersed by repetitions of max. 13 Familiarisation trials).

**Familiarisation – individual (6 trials, duration 15s per trial).** In these trials, an unmarked white block lay on the table directly in front of the agent. After addressing the infant, the agent would pick up the block and place it on the machine. The block would remain on the machine for 4 s, after which the agent would pick it up again and hold it in front of her, equidistant from the two sorting boxes. During the 4 s while the block was in contact with the surface of the machine, the machine produced an audio-visual effect in half of the trials and remained inactivated in the other half (order counterbalanced). After the agent had lifted the block off the machine and held it centrally, both of the sorting boxes lit up and a sound was played. After 2 s, the agent placed the block into one of the sorting boxes (orange or green, on the left or right side of the table). During placement, only the box into which the block was placed lit up a second time, and a second sound was played.

**Familiarisation – pairs (6 trials, duration 22s per trial).** As opposed to individual familiarisation trials, in these trials the blocks were retrieved from the sorting boxes and were marked with the corresponding colours of the box from which they were taken. Three types of pair trials were presented, involving 3 different pairs of blocks: 2 orange blocks; 2 green blocks; and 1 orange and 1 green block. When the two blocks were simultaneously

placed on the machine, an audio-visual effect was triggered in 2/3 of the trials, and not in 1/3 of the trials. The presence of the effect was consistent with the data in the individual trials: if the blocks that triggered the effect in individual trials were placed in the green box, then trials with 2 green blocks would likewise trigger the effect; whereas 2 orange blocks would not. Additionally, when one of each type of blocks was placed on the machine (one orange and one green) the audio-visual effect would always be triggered. After lifting the pairs of blocks off the machine, the agent would place them back into the sorting boxes, accompanied with the same 2 visual and audio cues as in *Familiarisation – individual* trials. The colour and side of the active blocks was fixed within each testing session but counter-balanced across the participants.

**Test trials (max. 16, duration 19s per trial, Fig. 1).** Test trials were split in two conditions (in quasi-randomized order); *Confounded* information trials, in which the unmarked block was paired with an 'active' block; and *Unconfounded* information trials, in which the unmarked block was paired with an 'inactive' block. In test trials, the agent paused while holding the blocks above the machine for 4 s prior to placing them on its surface. This created the *Anticipation* period of the trial, which was followed by the *Effect* period, during which the blocks were in contact with the machine. The machine activated in half of the *Unconfounded* trials and in all the *Confounded* trials. Finally, the test trials ended with a *Prediction* period: after lifting the blocks off the machine and hiding the known (active or inactive) block, the agent held the unmarked block centrally. As in *Familiarisation* trials, both sorting boxes lit up with the accompanying sound cue, marking the beginning of the *Prediction* period. In contrast to *Familiarisation* trials, in *Test* trials, the agent did not place the unmarked block in either of the sorting boxes but remained holding the block equidistant to the sorting boxes for 4 seconds. The second sound cue (associated with the agent placing the block into a box in familiarisation trials) was played 2 seconds prior to end of trial, but neither of the sorting boxes lit up for the second time. The presentation of *Test* trials was interspersed with repetitions of *Familiarisation* trials (max. 13; 6 individual, 7 pair) in order to reinforce the causal rule as well as infants' expectation of the objects being sorted into boxes at the end of the trials.

*Experiment 1* differed from the above in the following ways: 1) Lighting up of sorting boxes: Both boxes lighting up with an additional sound, prior to the sorting, was not present in *Experiment 1*. This was added for *Experiment 2*, in order to encourage saccades towards the sorting boxes during the *Prediction* period (which were not observed in *Experiment 1*). In *Familiarisation* trials of *Experiment 1*, only the box into which the block was sorted lit up with an accompanying sound, at the time when the block was placed into it. No box lit up in *Test* trials of *Experiment 1*. 2) *Familiarisation – pairs* videos ended with the agent lifting the blocks off the machine and did not show the agent sorting the blocks back into boxes. Therefore, the *Familiarisation – pairs* trials of *Experiment 1* were 9 s shorter than those of *Experiment 2*. 3) The audio-visual effect was triggered in the *Effect* period of all the *Test* trials (*Unconfounded* and *Confounded*) of *Experiment 1*. 4) Maximum number of *Test* trials in *Experiment 1* was 20.

## Measures

**EEG theta oscillations (Experiment 1 and 2).** Continuous EEG data was segmented around the beginning of the *Anticipation* period (4000 ms trial time) and visually inspected for artifacts, with noisy channels excluded and interpolated. The artifact free segments were subjected to time-frequency analysis. The epochs were imported into Matlab using the free toolbox EEGLAB (v. 2022.0) and re-referenced to average reference. Using 'Wtools', continuous wavelet transformation was applied and complex Morlet wavelets were computed for the frequency range 1–10 Hz with 1 Hz resolution. Epochs were averaged for each condition separately. To remove the distortion introduced by wavelet transform, 400 ms at each edge of the epochs were discarded. This resulted in 4600 ms long segments, including 600 ms before and 4000 ms after the onset of *Anticipation* period. The



average amplitude of the 600 ms before Anticipation period was used as baseline, subtracting it from the whole epoch at each frequency. Duration of baseline period was chosen to include approximately 2 cycles of the lowest frequency of interest (3 Hz).

**Gaze behaviour (Experiment 2).** Two main areas of interest (AOI) were defined for *Unconfounded* trials: Correct and Incorrect. Correct AOI corresponded to the box into which the block should be sorted (active box following an effect, inactive box following no effect). AOIs were same-sized and equidistant from the centre of the screen, surrounding the part of the scene where the sorting boxes (as well as the arms of the model when sorting objects) were positioned (see Fig. 1). **First look:** The first fixation within either of the AOIs (Correct or Incorrect) during *Prediction* period (15000–19000 ms) was extracted for each trial and assigned a score of 1 (Correct AOI), or -1 (Incorrect AOI). Trials with valid data but no fixations to either AOI were scored as 0. **Prediction Scores (PS)** were calculated by averaging the *First look* scores across the trials. **Differential looking score (DLS):** Cumulative duration of fixations within either of the box AOIs (Correct or Incorrect) during *Prediction* period were extracted. DLS was calculated [(looking time to Correct minus looking time to the Incorrect AOI) divided by (total looking time)] for each test trial and averaged across trials.

To control for low-level matching, *First looks* in *Confounded* trials were analysed analogously to *Unconfounded* trials, whereby a fixation within either of the AOIs during *Prediction* period was extracted for each trial and assigned a score of 1 (Matched AOI – active box), -1 (No-Match AOI – inactive box), or 0. **Matching Score** was calculated by averaging the *First look* scores across the *Confounded* trials.

### Inclusion criteria

Data was analysed for infants' attention to the screen (judged based on eyetracking or video recordings), and any trials in which the infant was not attending were excluded from analyses. To be included in the sample, participants had to see a minimum of 5 *Familiarisation* trials (one of each type; the 2 different individual object and the 3 different pair object trials) prior to presentation of *Test* trials. After exclusion for inattention, *Test* trials were visually screened for EEG artefacts and manually excluded for poor quality data. A minimum of 6 artefact free *Test* trials (3 trials per condition) was required for EEG data analysis. Infants in *Experiment 1* contributed an average of 8.06 valid EEG trials (*Unconfounded*,  $M = 4.0$  (range: 3–6); *Confounded*,  $M = 4.06$  (range: 3–5)); and infants in *Experiment 2* contributed on average 9.29 valid EEG trials (*Unconfounded*,  $M = 4.63$  (range: 3–8); *Confounded*,  $M = 4.67$  (range: 3–8)).

**Experiment 2, gaze data:** Two separate inclusion criteria were applied for gaze data measures. **First look:** trials were excluded if no fixation was detected anywhere on the screen during the *Prediction* period. **DLS:** trials were excluded if valid gaze data (at least one of the eyes is reliably tracked) was collected in less than 50% of the *Prediction* period duration. On average, infants contributed 11.75 valid eyetracking trials (*First look* measure: *Unconfounded*,  $M = 5.96$  (range: 3–8); *Confounded*,  $M = 5.79$  (range: 3–8); **DLS** measure: *Unconfounded*,  $M = 5.38$  (range: 2–8); *Confounded*,  $M = 5.41$  (range: 1–8)). All datasets included in *Experiment 2* analyses contained minimum 6 test trials with valid EEG, of which 4 had to additionally contain valid *First look* data. Note, however, that the reported analyses were not limited to trials including valid data in both EEG and eyetracking measures; EEG results include all valid EEG trials (regardless of valid gaze data) and the reverse.

## Results

### Evaluating informativeness of evidence - EEG theta oscillations

Continuous wavelet analysis of artifact-free epochs, in the 1–10 Hz range, was performed on a cluster of 9 electrodes over the central-frontal scalp locations, where modulations of theta activity were predicted based on previous findings in adults and infants<sup>25,26,35</sup>. Baseline corrected data were analyzed for amplitude in the 3–5 Hz theta frequency range, during the *Anticipation* period. A  $2 \times 2$  repeated measures ANOVA with *Experiment*

(1, 2) as between subject variable, and Condition (*Confounded*, *Unconfounded*) as within subject variable revealed a main effect of Condition ( $F(64,1) = 8.585, p = 0.005, \mu_p^2 = 0.118$ ). There was no significant main effect or interactions involving the between-subject variable *Experiment*. As predicted, the amplitude of theta (3–5 Hz) oscillatory activity during the *Anticipation* period was significantly greater in *Unconfounded* than *Confounded* condition ( $M_{diff} = 0.223$ ;  $SE = 0.076$ ;  $t = 2.930$ ;  $p_{holm} = 0.005$ , Cohen's  $d = 0.384$ , 95% CI [0.116 0.6524]; Fig. 2, combining data from *Experiment 1* and 2).

### Inferring causal roles - Gaze behavior

Data of *Experiment 2* ( $N = 32$ ) was additionally analyzed for infants' gaze behavior during the *Prediction* period of the trial. This analysis was focused on *Unconfounded* condition, in which infants had sufficient information to form a causal inference and therefore to predict into which sorting box the unmarked block will be placed. We investigated whether infants made predictive saccades towards the sorting boxes and computed *Prediction Scores* (PS = averaged *First look* scores (see *Measures*) across trials). A PS = 0 on this measure indicates that an infant did not make any predictive saccades or was equally likely to first look at either of the sorting boxes across the *Test* trials; a score of PS > 0 would indicate that an infant made correct predictions on more than 50% of the trials; and PS < 0 indicates an infant made more incorrect than correct predictions. *Predictive scores* revealed no statistically significant evidence that, as a group, infants performed above chance ( $t(31) = 0.338, p = 0.738$ , Cohen's  $d = 0.06$ , 95% CI [-0.288 0.406]). Analysis of the cumulative duration of infants' fixations within the areas surrounding the sorting boxes (*Differential looking score*, see *Measures*) likewise did not indicate any systematic gaze behavior towards either of the sorting boxes during the *Prediction* period ( $t(31) = 1.451, p = 0.157$ , Cohen's  $d = 0.257$ , 95% CI [-0.098 0.607]).

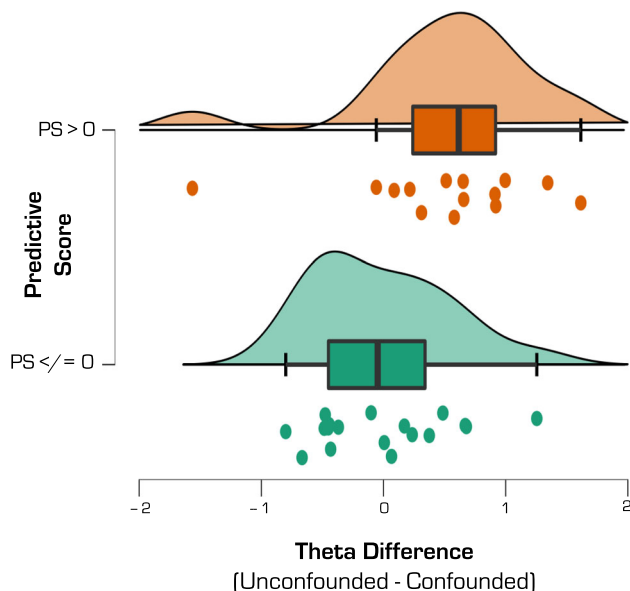
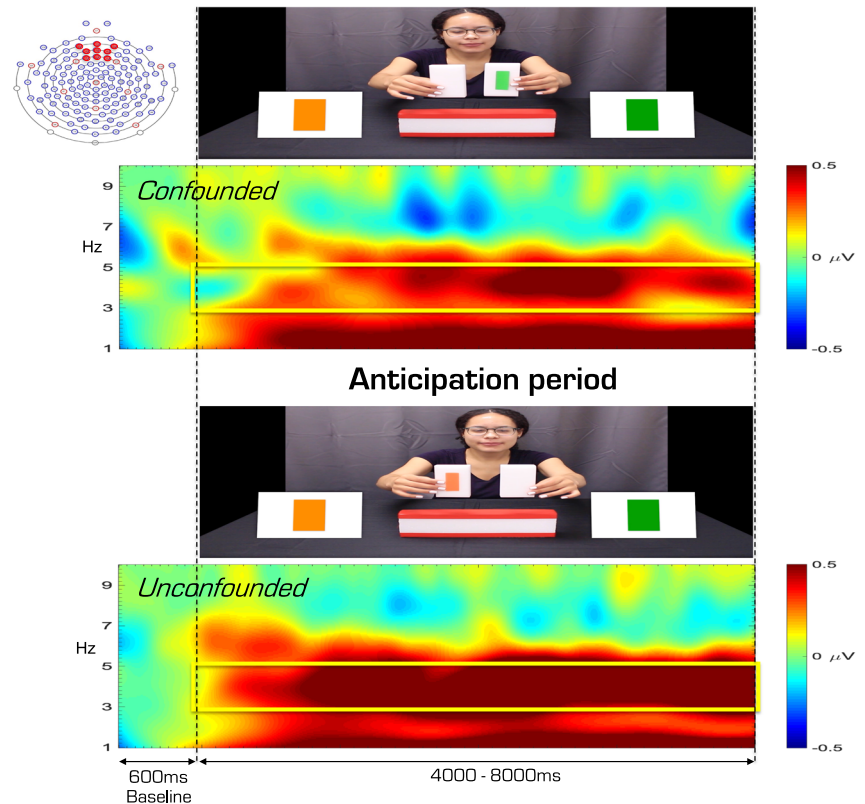
### Relating evidence evaluation and ability to infer causal roles

Crucially, we were interested in whether infants, who made correct predictions on more than 50% of the trials, differed from the infants, who did not. Specifically, we hypothesized that infants' ability to evaluate the informativeness of expected evidence (as reflected in the difference in theta oscillations between *Unconfounded* and *Confounded* conditions during the *Anticipation* period) may underlie the ability to correctly infer the observed causal structure and therefore make correct predictions. To investigate this, we split the sample of *Experiment 2* into infants with *Prediction Scores* larger than 0 ( $N = 15$ ), and those with PS = / < 0 ( $N = 17$ ) and compared their sensitivity to confounded information (difference in amplitude of theta oscillations between *Unconfounded* and *Confounded* conditions). A one-way ANOVA revealed a significant difference between infants with positive *Prediction Scores* and infants with PS = / < 0 ( $F(30,1) = 4.556, p = 0.041, \mu_p^2 = 0.132$ ). Infants, who made mostly correct predictive saccades during the *Prediction* period, showed greater differentiation between confounded and unconfounded information during the *Anticipation* period ( $M_{diff} = 0.496$ ;  $SE = 0.233$ ;  $t = 2.135$ ;  $p_{tukey} = 0.041$ , Cohen's  $d = 0.756$ , 95% CI [0.009 1.503]; Fig. 3). No statistically significant correlations were found between anticipatory theta difference and PS ( $r = 0.273, p = 0.130$ ) or DLS measures ( $r = 0.168, p = 0.359$ ).

### Controlling for low-level explanations

As per the design of *Experiment 2*, the audiovisual effect was triggered in 50% of the *Unconfounded* trials, and not triggered in the rest. This meant that the causal role that could be attributed to the novel object (the novel object must be active or inactive) mapped onto the presence or absence of the triggered effect. In trials where the effect was triggered, the novel object should be inferred to be active, and the reverse for trials where no effect was triggered. Therefore, it is possible that infants' predictive saccades towards the sorting boxes in *Unconfounded* trials did not reflect causal inferences regarding the novel objects but were based on a simpler low-level mechanism of matching the presence/absence of the effect with the corresponding active/inactive sorting boxes. To control for this possibility, we

**Fig. 2 | EEG results.** Data presented is the averaged neural activity across the marked cluster of 9 electrodes over the central-frontal recording site. Time-frequency plots depict baseline-corrected data during *Anticipation* period, averaged across *Test* trials, combining *Experiment 1* and 2 ( $N = 66$ ). Baseline period immediately precedes analysis period and is recorded while participants observe the agent picking up the selected blocks and bringing them above the machine (600 ms duration). Amplitude of oscillations between 1 and 10 Hz is presented separately for *Unconfounded* (top) and *Confounded* (bottom) conditions. Yellow frames mark the frequency range (3–5 Hz) and time period (4000–8000 ms trial time) used for analysis. Significantly more theta activity is observed during the anticipation window of the *Unconfounded* trials.



**Fig. 3 | Relationship between Predictive Scores and Theta activity.** Differences in amplitude of theta oscillations (*Unconfounded* – *Confounded* condition) during *Anticipation* period, plotted separately for infants who made mostly correct anticipatory saccades during the *Prediction* period (*Predictive Score* > 0;  $N = 15$ , top), and infants who did not ( $PS = / < 0$ ;  $N = 17$ , bottom). The subset of infants who made correct predictions ( $PS > 0$ ), suggesting that they had correctly inferred the causal structure, showed a more pronounced differentiation between unconfounded and confounded evidence, as reflected in a positive difference in theta activity. Raincloud Plot shows the data as data distribution plots, individual data points (representing individual participants), and boxplots (representing median (inner line), 25th and 75th percentile (box edges) and minimum and maximum values before the upper/lower fence (arms)).

additionally analyzed infants' predictive saccades in *Confounded* trials. If it were the case that infants' predictions reflect low-level matching, it should follow that infants would consistently look to the active box in *Confounded* trials, in which the effect was always triggered. *Matching Scores* were computed analogously to *Prediction scores* ( $MS$  = averaged *First look* scores during *Prediction* period of *Confounded* trials).  $MS$  indicates whether infants made no predictions or were equally likely to first look at either of the sorting boxes across the trials ( $MS = 0$ ), matched the triggering of the effect to the active box on more than 50% of the trials ( $MS > 0$ ), or were more likely to match the effect with the inactive sorting box ( $MS < 0$ ). The results revealed no evidence of low-level matching: infants' predictions in *Confounded* trials did not differ from chance ( $t(31) = 1.299$ ,  $p = 0.203$ , Cohen's  $d = 0.230$ , 95% CI  $[-0.123, 0.579]$ ). Importantly, the infants who made on average correct predictions in *Unconfounded* trials ( $PS > 0$ ) had on average a negative *Matching Score* ( $M = -0.121$ ,  $SD = 0.314$ ) and were significantly less likely to do low-level matching in *Confounded* trials than were infants with  $PS = / < 0$  ( $M = 0.329$ ,  $SD = 0.570$ ; one-way ANOVA:  $F(30,1) = 7.330$ ,  $p = 0.011$ ,  $\mu_p^2 = 0.196$ ).

## Discussion

This study investigated the sensitivity to the informativeness of evidence in 16-month-old infants, and how such sensitivity may relate to infants' ability to draw correct causal inferences and predict unfolding events. Using concurrent EEG and eye tracking recordings, the data revealed significantly more theta oscillatory activity when infants were expecting to see unconfounded compared to confounded evidence, suggesting heightened cognitive engagement in anticipation of informative evidence. Crucially, this difference was more pronounced in the subset of infants who later made correct predictions, suggesting that they had inferred the causal structure based on the evidence presented. These results provide evidence that (some) infants at 16 months can evaluate the informativeness of evidence even before it is given and selectively engage with information that is unconfounded. We propose that theta activity in this paradigm reflected an expectation of causally relevant information, and that this anticipatory

cognitive engagement facilitated the encoding of relevant causal information, which in turn facilitated infants forming correct causal inferences and predictions. Moreover, we argue that such selectivity reflects an optimal use of one's cognitive resources, can prevent potential unjustified conclusions formed based on confounded evidence, and can support infants' rapid learning about the causally opaque world.

In line with existing evidence that theta oscillations can be observed in situations where infants can expect to receive information<sup>25</sup>, are predictive of infant learning<sup>26</sup>, and are triggered by contexts of infant learning (e.g. violation of expectation<sup>28,29</sup>), we interpret the observed difference in theta activity in the current paradigm as infants selectively engaging their cognitive resources in situations where evidence that is causally informative is to be gained. As theta activity has also been associated with reward processing<sup>36</sup>, it is important to note that the observed pattern of results could not be explained by infants anticipating the triggering of the effect and finding the effect rewarding. In *Experiment 2*, only half of the *Unconfounded* test trials resulted in the effect being triggered, whereas all *Confounded* trials produced the effect. An anticipation of the reward (triggering of the effect) would thus predict the opposite pattern of results.

Current results complement prior findings demonstrating that infants are good at judging when effort should be invested in a task<sup>37</sup> when persistence might pay off<sup>38</sup>, when uncertainty warrants evidence accumulation<sup>4</sup>, or when one should seek help to avoid making mistakes<sup>39</sup>. Here, we demonstrate that infants not only rationally deploy *behavioural* effort depending on task demands but do so also with their limited *cognitive* capacities. Our data also suggests that even young infants, while they still have limited ability to intervene on the world themselves, already appear motivated to gain explanatory causal information. Seeking explanations and deliberately uncovering the underlying causal mechanisms of phenomena that we observe is likely a uniquely human capacity. Many animals explore their environments, seek information that can lead to food or mates; predict events that are of importance for their survival; and demonstrate impressive abilities in selecting and designing tools<sup>40</sup>. However, no evidence to date suggests that non-human animals deliberately seek causal explanations, or strive to understand the causal structures in their environment<sup>41</sup>. Although causal understanding may not be necessary for the evolution of human technology<sup>42</sup>, explanation seeking curiosity is an important driver of learning as it has the power to reveal the unobservable and causal structures of the world and thus supports generalizable knowledge<sup>43</sup>. Not only have children been shown to distinguish genuine causes from spurious correlations<sup>44</sup>; when given ambiguous evidence, preschoolers spontaneously select and design informative interventions to reveal the causal relationships between the objects they are engaged with<sup>45</sup>. While some work has suggested that infants may be able to use conditional independence to form causal predictions by 8-months<sup>46</sup>, limited prior research has suggested that causal explanation seeking may be present already in infancy. One such study has shown that 11-month-old infants' preference for exploring an object, which has violated their expectations (by e.g. seemingly passing through a solid wall), disappears if an explanation is provided for the surprising event (the wall is revealed to have a hole through which the object could pass)<sup>47</sup>. While infants in the same paradigm would otherwise choose to explore the surprising object and even appear to test its surprising properties<sup>6</sup>, the absence of such behavior, when an explanation was provided, suggests that infants' exploration was motivated by the desire to explain the surprising events. The results of the current study compliment these findings and demonstrate that not only are infants motivated to explain past surprising events, but they also selectively engage their cognitive resources and proactively direct their attention to situations in which explanatory causal information can be gained.

## Limitations

A possible alternative interpretation of the observed differences could be that the heightened theta in *Unconfounded* trials reflected not infants' expectation of relevant causal information, but rather infants' uncertainty regarding the upcoming triggering of the effect. While theta activity has not been systematically investigated in relation to uncertainty in infants, some

adult research does suggest a relationship between uncertainty tracking and frontal theta activity<sup>48</sup>. Below, we spell out why we believe that heightened theta activity in *Unconfounded* trials is not likely to reflect a state of uncertainty in our task.

In *Experiment 1*, all *Test* trials, *Confounded* and *Unconfounded*, resulted in the effect being triggered. If theta activity reflected uncertainty, and infants had not formed any expectations about the upcoming information based on causal reasoning, then uncertainty-reflecting-theta activity should be high at the beginning of the experiment and diminish with the accumulating evidence over the course of the trials. In absence of any causal reasoning, this should apply equally to both *Confounded* and *Unconfounded* trials, as the evidence (triggering of the effect regardless of selected blocks) is the same and needs to be accumulated in both conditions alike. Therefore, we would expect no difference in theta activity between the conditions of *Experiment 1*. This explanation is inconsistent with observed data and can therefore be ruled out. Alternatively, it is possible that infants in *Experiment 1* paid attention to whether the information is confounded, but heightened theta activity in *Unconfounded* trials did not reflect heightened cognitive engagement in expectation of relevant causal information, but instead reflected a higher level of uncertainty associated with the expected effect. In this case, infants should start with higher uncertainty about the effect in *Unconfounded* trials (based on causal inferences, the likelihood of the effect should be 50%) compared to *Confounded* trials (likelihood of effect is always 100%). As infants accumulate information over the course of the trials, their expectations regarding the likelihood of seeing the effect in *Unconfounded* trials may change (based on the evidence observed, the expectation of seeing the effect 50% of the time might shift towards 100% of the time). Therefore, it would be expected that theta activity in *Unconfounded* trials of *Experiment 1* would decrease over the course of the experiment, as the uncertainty decreases with accumulating (counter-)evidence. Due to limited number of trials, we could not perform analyses that would exclude this possibility. However, even if heightened theta in *Unconfounded* trials of *Experiment 1* reflected (initial) uncertainty about the upcoming effect, this uncertainty nonetheless necessitates that infants had computed the likelihood of the effect based on reasoning about the causal roles of the blocks to begin with. Without such computations, we find no explanation for the observed difference between the conditions.

Therefore, we suggest that infants must have been doing something rather sophisticated. Prior research has already shown that infants can map certain features of the stimuli, such as the identity of the informant<sup>25</sup>, or type of shape<sup>20</sup> to the upcoming events in order to form selective expectations of receiving information. To form different expectations between confounded and unconfounded trials in the current study, we argue that infants had to, at the minimum, combine their knowledge of multiple individual components (the different blocks and their known and unknown causal roles) to form an expectation of whether the particular combination of the components is likely to result in the effect being triggered.

Finally, our main argument for interpreting heightened theta in *Unconfounded* trials as reflecting expectation of causally relevant information (and not a state of uncertainty) comes from the combined EEG and eyetracking data of *Experiment 2*. Given the difference in the likelihood of the effect being triggered in the two conditions of *Experiment 2*, the difference in uncertainty-reflecting-theta could be expected to increase over the course of *Experiment 2* (with accumulating evidence for 100 vs. 50% likelihood of the effect in the two conditions). While this is plausible and could happen with or without the infants making any computations regarding the evidence being confounded or not, there is no readily available explanation for why a state of heightened uncertainty when facing unconfounded evidence would lead to forming correct causal inferences and predictions. In contrast, studies have shown that frontal theta activity in expectation of stimulus can reflect a relevance-signalling mechanism and is involved in control over the selective encoding of relevant events<sup>49</sup>, as well as that heightened theta prior to encoding predicts the success of encoding<sup>35</sup>. The current results have shown that the infants who succeeded in predicting the outcomes of the events showed more pronounced difference in theta



activity prior to viewing the evidence compared to infants who showed no evidence of making correct predictions. We therefore conclude that the most parsimonious interpretation of all our combined results is that heightened anticipatory theta activity in this study reflected infants' selective cognitive engagement prior to viewing unconfounded evidence, and that such heightened activity plausibly facilitated the processing of the observed information and thereby resulted in increased likelihood of forming correct inferences and predictions.

## Conclusion

Leveraging the strength of our approach, combining neural activity prior to presentation of evidence, and behavioral predictions that follow the neural activity, we provided not only evidence that infants recognise situations in which informative evidence can be gained, but also that this sensitivity is possibly what supports infants forming correct inferences regarding observed causal structures. Thus, the possibly uniquely human motivation to seek causal explanations, combined with the ability to evaluate the informativeness of observed evidence and use it to form causal inferences and predictions, may well be the developmental precursor or prerequisite of the impressive abilities of humans to rapidly acquire generalizable knowledge and develop causal theories of the world<sup>50</sup>.

## Data availability

According to GDPR, raw eye-tracking and raw EEG data cannot be fully anonymised, and therefore cannot be made publicly available under the restrictions of the study's ethical approval and informed consent. Raw data can be shared with individual researchers or institutions upon request addressed to the corresponding author and pending a data-sharing agreement. Numerical data underlying the Figures is available on: [https://osf.io/zxcn4/?view\\_only=743b49cd78c4416e861f952b0e53c148](https://osf.io/zxcn4/?view_only=743b49cd78c4416e861f952b0e53c148).

## Code availability

Experiment was conducted using SR Research Experiment Builder. Eye-tracking data was recorded and extracted from SR Research Data Viewer, and EEG data was recorded using Magstim EGI Geodesic EEG System, and preprocessed in software Netstation 5.4.1.1. Further processing of EEG data was conducted in Matlab (R2022a) toolbox EEGLAB (v. 2022.0) and Wtools. Figure 3, as well as statistical tests, were generated using freely available software JASP. The statistical analyses code is available on: [https://osf.io/zxcn4/?view\\_only=743b49cd78c4416e861f952b0e53c148](https://osf.io/zxcn4/?view_only=743b49cd78c4416e861f952b0e53c148).

Received: 1 March 2024; Accepted: 13 August 2024;

Published online: 20 August 2024

## References

1. Csibra, G. & Gergely, G. Natural pedagogy as evolutionary adaptation. *Philos. Trans. R. Soc. Lond. Ser. B, Biol. Sci.* **366**, 1149–1157 (2011).
2. Choi, D., Batterink, L. J., Black, A. K., Paller, K. A. & Werker, J. F. Preverbal Infants Discover Statistical Word Patterns at Similar Rates as Adults: Evidence From Neural Entrainment. *Psychol. Sci.* **31**, 1161–1173 (2020).
3. Poli, F., Meyer, M., Mars, R. B. & Hunnius, S. Contributions of expected learning progress and perceptual novelty to curiosity-driven exploration. *Cognition* **225**, 105119 (2022).
4. Leckey, S. et al. Response latencies and eye gaze provide insight on how toddlers gather evidence under uncertainty. *Nat. Hum. Behav.* **4**, 928–936 (2020).
5. Fantz, R. L. Visual Experience in Infants: Decreased Attention to Familiar Patterns Relative to Novel Ones. *Science* **146**, 668–670 (1964).
6. Stahl, A. E. & Feigenson, L. Observing the unexpected enhances infants' learning and exploration. *Science* **348**, 91–94 (2015).
7. Wynn, K. Addition and subtraction by human infants. *Nature* **358**, 749–750 (1992).
8. Kidd, C., Piantadosi, S. T. & Aslin, R. N. The Goldilocks effect: human infants allocate attention to visual sequences that are neither too simple nor too complex. *PLoS One* **7**, e36399 (2012).
9. Twomey, K. E. & Westermann, G. Curiosity-based learning in infants: a neurocomputational approach. *Dev. Sci.* **21**, e12629 (2018).
10. Johnson, M. H., Dziurawiec, S., Ellis, H. & Morton, J. Newborns' preferential tracking of face-like stimuli and its subsequent decline. *Cognition* **40**, 1–19 (1991).
11. Mueller, J. L., Friederici, A. D. & Männel, C. Auditory perception at the root of language learning. *Proc. Natl Acad. Sci. USA* **109**, 15953–15958 (2012).
12. Farroni, T., Csibra, G., Simion, F. & Johnson, M. H. Eye contact detection in humans from birth. *Proc. Natl Acad. Sci. USA* **99**, 9602–9605 (2002).
13. Cooper, R. P. & Aslin, R. N. Preferences for Infant-directed Speech in the First Month after Birth. *Child Dev.* **61**, 1584–1595 (1990).
14. Csibra, G. & Gergely, G. Natural pedagogy. *Trends Cogn. Sci.* **13**, 148–153 (2009).
15. Bonawitz, E. & Shafto, P. Computational models of development, social influences. *Curr. Opin. Behav. Sci.* **7**, 95–100 (2016).
16. Manea, V., Kampis, D., Grosse Wiesmann, C., Revencu, B. & Southgate, V. An initial but receding altercentric bias in preverbal infants' memory. *Proc. Royal Soc. B* **290**, 1–11 (2023).
17. Begus, K. & Southgate, V. Infant pointing serves an interrogative function. *Dev. Sci.* **1–8** <https://doi.org/10.1111/j.1467-7687.2012.01160.x> (2012).
18. Tummeltshammer, K. S., Wu, R., Sobel, D. M. & Kirkham, N. Z. Infants Track the Reliability of Potential Informants. *Psychol Sci* <https://doi.org/10.1177/0956797614540178> (2014).
19. Bazhydai, M., Westermann, G. & Parise, E. "I don't know but I know who to ask": 12-month-olds actively seek information from knowledgeable adults. *Dev. Sci.* **23**, e12938 (2020).
20. Ghilardi, T., Poli, F., Meyer, M., Colizoli, O. & Hunnius, S. Early roots of information-seeking: Infants predict and generalize the value of information. *Elife* **13**, 1–29 (2024).
21. Poli, F., Serino, G., Mars, R. B. & Hunnius, S. Infants tailor their attention to maximize learning. *Sci. Adv.* **6**, 5053–5076 (2020).
22. Schulz, L. E. & Bonawitz, E. B. Serious Fun: Preschoolers Engage in More Exploratory Play When Evidence Is Confounded. *Dev. Psychol.* **43**, 1045–1050 (2007).
23. Rutishauser, U., Ross, I. B., Mamelak, A. N. & Schuman, E. M. Human memory strength is predicted by theta-frequency phase-locking of single neurons. *Nature* **464**, 903–907 (2010).
24. Orekhova, E. V. & Stroganova, T. a & Posikera, I. N. Theta synchronization during sustained anticipatory attention in infants over the second half of the first year of life. *Int. J. Psychophysiol.* **32**, 151–172 (1999).
25. Begus, K., Gliga, T. & Southgate, V. Infants' preferences for native speakers are associated with an expectation of information. *Proc. Natl. Acad. Sci.* **201603261** <https://doi.org/10.1073/pnas.1603261113> (2016).
26. Begus, K., Southgate, V. & Gliga, T. Neural mechanisms of infant learning: differences in frontal theta activity during object exploration modulate subsequent object recognition. *Biol. Lett.* **11**, 20150041–20150041 (2015).
27. Begus, K. & Bonawitz, E. The rhythm of learning: Theta oscillations as an index of active learning in infancy. *Dev. Cogn. Neurosci.* **45**, 100810 (2020).
28. Berger, A., Tzur, G. & Posner, M. I. Infant brains detect arithmetic errors. *Proc. Natl Acad. Sci. USA* **103**, 12649–12653 (2006).
29. Köster, M., Langeloh, M., Michel, C. & Hoehl, S. Young infants process prediction errors at the theta rhythm. *Neuroimage* **236**, 118074 (2021).



30. Wass, S. V. et al. Parental neural responsivity to infants' visual attention: How mature brains influence immature brains during social interaction. *PLoS Biol.* **16**, e2006328 (2018).
31. Xie, W., Mallin, B. M. & Richards, J. E. Development of infant sustained attention and its relation to EEG oscillations: an EEG and cortical source analysis study. *Dev. Sci.* **21**, e12562 (2018).
32. Jones, E. J. H., Venema, K., Lowy, R., Earl, R. K. & Webb, S. J. Developmental changes in infant brain activity during naturalistic social experiences. *Dev. Psychobiol.* **57**, 842–853 (2015).
33. Fell, J. et al. Medial temporal theta/alpha power enhancement precedes successful memory encoding: evidence based on intracranial EEG. *J. Neurosci.* **31**, 5392–5397 (2011).
34. Guderian, S., Schott, B. H., Richardson-Klavehn, A. & Düzel, E. Medial temporal theta state before an event predicts episodic encoding success in humans. *Proc. Natl Acad. Sci. USA* **106**, 5365–5370 (2009).
35. Gruber, M. J., Watrous, A. J., Ekstrom, A. D., Ranganath, C. & Otten, L. J. Expected reward modulates encoding-related theta activity before an event. *Neuroimage* **64**, 68–74 (2013).
36. Christie, G. J. & Tata, M. S. Right frontal cortex generates reward-related theta-band oscillatory activity. *Neuroimage* **48**, 415–422 (2009).
37. Lucca, K., Horton, R. & Sommerville, J. A. Infants rationally decide when and how to deploy effort. *Nat. Hum. Behav.* **2020** *4*, 372–379 (2020).
38. Leonard, J. A., Lee, Y. & Schulz, L. E. Infants make more attempts to achieve a goal when they see adults persist. *Science* (1979) **357**, 1290–1294 (2017).
39. Goupil, L., Romand-monniere, M. & Kouider, S. Infants ask for help when they know they don't know. **2016**, (2016).
40. Manrique, H. M. & Call, J. Spontaneous use of tools as straws in great apes. *Anim. Cogn.* **14**, 213–226 (2011).
41. Völter, C. J. & Call, J. Causal and inferential reasoning in animals. *APA handbook of comparative psychology: Perception, learning, and cognition*. 643–671 <https://doi.org/10.1037/0000012-029> (2017).
42. Derex, M., Bonnefon, J. F., Boyd, R. & Mesoudi, A. Causal understanding is not necessary for the improvement of culturally evolving technology. *Nat. Hum. Behav.* **3**, 446–452 (2019).
43. Liquin, E. G. & Lombrozo, T. Explanation-seeking curiosity in childhood. *Curr. Opin. Behav. Sci.* **35**, 14–20 (2020).
44. Gopnik, A., Sobel, D. M., Schulz, L. E. & Glymour, C. Causal learning mechanisms in very young children: two-, three-, and four-year-olds infer causal relations from patterns of variation and covariation. *Dev. Psychol.* **37**, 620–629 (2001).
45. Cook, C., Goodman, N. D. & Schulz, L. E. Where science starts: Spontaneous experiments in preschoolers' exploratory play. *Cognition* **120**, 341–349 (2011).
46. Sobel, D. M. & Kirkham, N. Z. Blickets and babies: The development of causal reasoning in toddlers and infants. *Dev. Psychol.* **42**, 1103–1115 (2006).
47. Perez, J. & Feigenson, L. Violations of expectation trigger infants to search for explanations. *Cognition* **218**, 104942 (2022).
48. Cavanagh, J. F., Figueroa, C. M., Cohen, M. X. & Frank, M. J. Frontal theta reflects uncertainty and unexpectedness during exploration and exploitation. *Cereb. Cortex* **22**, 2575–2586 (2012).
49. Jarovi, J., Volle, J., Yu, X., Guan, L. & Takehara-Nishiuchi, K. Prefrontal Theta Oscillations Promote Selective Encoding of Behaviorally Relevant Events. *eNeuro* **5**, 1–16 (2018).
50. Gopnik, A. & Schulz, L. Mechanisms of theory formation in young children. *Trends Cogn. Sci.* **8**, 371–377 (2004).

## Acknowledgements

This research was funded by McDonnell Foundation and National Science Foundation (Grant No. SES-1627971) awarded to Elizabeth Bonawitz. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript. We thank Victoria Southgate and Centre for Early Childhood Cognition at University of Copenhagen for generously hosting the first author and the data collection during the COVID-19 pandemic, Kristina Thygesen for her assistance with data collection, Dimitrios Tsakiris and Sabrina Junk for their assistance with data analysis, and families that took part in the study. We thank Carla Macias for help in creating the video stimuli. The authors have received consent for publication of included images.

## Author contributions

Katarina Begus designed the study, collected and analysed the data, and co-wrote the manuscript. Elizabeth Bonawitz provided the funding, supervised the design of the study, and co-wrote the manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s44271-024-00131-3>.

**Correspondence** and requests for materials should be addressed to Katarina Begus.

**Peer review information** *Communications psychology* thanks the anonymous reviewers for their contribution to the peer review of this work. Primary Handling Editors: Saloni Krishnan and Jennifer Bellington. A peer review file is available.

**Reprints and permissions information** is available at <http://www.nature.com/reprints>

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024